

02

LECTURE

HEALTH CARE

MT 1, Second Semester LECTURE 02 - Ms. Guillerma L. Lim, RMT

HUMAN BEHAVIOR AND COMMUNITY HEALTH EDUCATION

TOPIC OVERVIEW

I. Behavior

- A. Classifications of Human Behavior
- B. Views on Human Behavior

II. Biological Bases of Behavior

- A. Control Systems
- B. Endocrine Systems
- C. The Brain

III. Health Behavior

- A. Risk and Health
- B. Major Family Health Risks
- C. Behavioral (Lifestyle) Risks
- D. Tobacco & Health Risks
- E. Alcohol Consumption & Health Risks
- F. Caffeine Intake

IV. Models/Theories of Human Behavior

- A. Health Belief Model
- B. Transtheoretical/Change Model
- C. Theory of Planned Behavior/Reasoned Action
- D. Social Cognitive Theory

V. Community Health Education

- A. Skills of the Effective Educator
- B. Use of Technology in Health Education

VI. Community Health Workers (Barangay Health Workers)

- A. Two-Way Referral System
- B. Diagnostic Stages/Phases of Educational Planning for Community Health

LEARNING OBJECTIVES

1. To be able to define behavior
2. To differentiate the different classifications, views, and biological bases on human behavior
3. To know the different major health and lifestyle risks
4. To know and understand the different theories/models of human behavior
5. To be able to identify the diagnostic stages/phases of educational planning for community health

I. BEHAVIOR

Behavior

- Externally visible activity of an animal, in which a coordinated pattern of sensory, motor, and associated neural activity responds to changing external or internal conditions." (Beck, et. al, 1981)
- A response to external and internal stimuli following the integration of (1) sensory, (2) neural, (3) endocrine, and (4) effector components.
- "Behavior has a genetic basis; hence is subject to natural selection, and it can be commonly modified through experience." (Starr and Taggart, 1992)
- "Behavior is a Observable activity of an organism; anything an organism does that

involves action and/or response to stimulation." (Wallace et al., 1991)

- In simpler terms, behavior is how someone acts and a response to things that are happening. Internally, thoughts and feelings while externally
- refer to the relations and environment around us. It is what a person does to make something happen. It is what a person does to make something change or it is what a person does to keep things the same.

A. CLASSIFICATIONS OF HUMAN BEHAVIOR

<u>MOLECULAR</u> <i>unconscious</i>	<u>MOLAR</u> <i>conscious</i>
• Actions are done by instinct • Behavior that occurs without thinking (unconscious behavior) • Also called the "reflex action" • e.g. A man hits a stick on another person's eyes and that person has to close his eyes in response.	• Actions are done with reason • One is aware of his actions (conscious behavior) • e.g. A man counterattacks in revenge.
<u>OVERT</u> <i>visible</i>	<u>COVERT</u> <i>invisible</i>
• Behavior that is being manifested or physically observable • e.g. A man playing basketball.	• Behavior that is not visible or observable • e.g. A man is thinking or a person breathing..
<u>VOLUNTARY</u> <i>controlled</i>	<u>INVOLUNTARY</u> <i>uncontrolled</i>

• Actions are with one's desire • Done with man's full will and discretion (voluntary) • e.g. speaking, writing	• Behavior that occurs naturally or actions that are done automatically (involuntary) • e.g. A man is taking oxygen and giving up carbon dioxide.
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Table 1. The Difference between the Classifications of Human Behavior

B. VIEWS ON HUMAN BEHAVIOR

Biological Viewpoint

Psychiatrist's Point of View

- Everything you think and feel is controlled by electrical and chemical activity in your brain and the rest of your body.
- To these theorists, the body controls the mind and not the other way around.

Intrapsychic Viewpoint

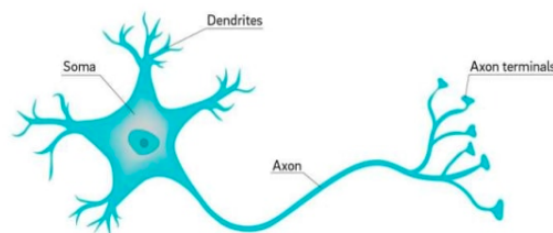
Psychologist's Point of View

- The psychologists try to explain the behaviors they've measured in terms of mental processes such as perceptions, motives, values, attitudes, and memories.
- The mind dominates most bodily activities
- This viewpoint is opposite to the biological viewpoint
- e.g. The psychologist will try to teach a depressed person to stop thinking about himself/herself in negative ways and encourage him/her to set realistic positive goals.

Social/Behavioral Viewpoint

- One's behavior can be explained in terms of their physical, social, and cultural environment.
- One is not an island unto oneself; he or she grew up around people who help determine his or her ideas, values, speech, and behaviors.
- E.g. A person who is depressed and was caused by a broken relationship, making her feel unspecial because she was built up by a father who gave her special attention and special treatment

II. BIOLOGICAL BASIS OF BEHAVIOR



Neuron

- It is a nerve cell that is used for communicating
- All 3 parts are necessary for transmitting necessary messages through your body
- Without these messengers you could not perform the simplest action or think a single thought

Nerve

- A large bundle of axons from many neurons

PARTS	FUNCTION
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Dendrite	• The front end or input side of a neuron composed of a network of fibers • Receives info from other nerve cells
Soma	• Main part of the body of nerve cell • Receives inputs from other neurons • Neuron's "housekeeper" • Most complex chem reactions in • Cellular metabolism happens here (keeps cell healthy)
Axon	• Action end or output system of the neuron • Stretches out from the soma like a branching telephone cable • At the end of its cable are tiny fibers (axon terminals) that release chemicals that affect the dendrites and cell bodies of nearby neurons, or muscles and glands

Table 2. Three Parts of a Neuron (aka the Messengers)

TYPE	FUNCTION
Sensory Neurons <i>afferent neurons</i>	• Involved in relaying information from your (1) eyes, (2) ears, and (3) other sensory receptors, then to the brain
Motor Neurons	• Involved in relaying

<i>efferent neurons</i>	messages from your brain to your muscles and gland
Interneuron	- Convey impulses from one neuron to another in the brain and spinal cord. - Most of brain's neural activity is carried on by interneurons

Table 4. Three Types of Neurons

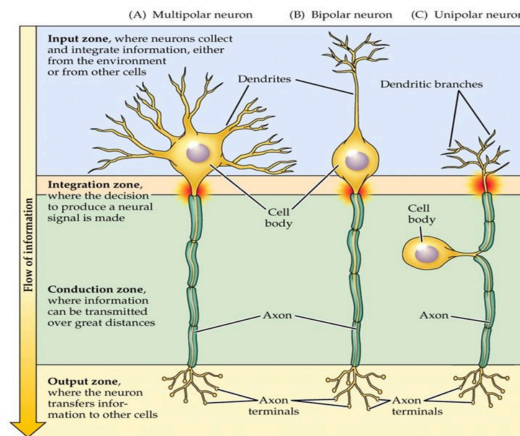


Figure 2. Classification of Neurons

A. CONTROL SYSTEMS

Central Nervous System (CNS)

- Consists of the brain and spinal cord

Peripheral Nervous System (PNS)

- Consists mainly of nerves connecting the central nervous system to the sense receptors as well as to the muscles and glands
- Somatic Nervous System**

- Controls the voluntary muscles
- Autonomic Nervous System (ANS)**
 - Controls the normal body functions (involuntary)
 - e.g. breathing, pumping out of blood from the heart

Autonomic Nervous System (ANS)

- Sympathetic Nervous System (SNS)**
 - Responsible for the body's flight and flight activities.
 - Speeds up bodily activities such as heart rate, breathing, and sweating
 - e.g. whenever you encounter an emergency, or something enrages you, you are frightened
 - Controls orgasm and ejaculation
- Parasympathetic Nervous System (PNS)**
 - Promotes breathing, digestion and secretion
 - Turns off or slows down most activities automatically aroused by the sympathetic system
 - Conserves or builds up your body's resources
 - Controls nipple erection (in females) and penis erection (in males)

Synapse

- general area where two neurons come in contact with each other
- A neuron **relays its message** or can communicate with another neuron across a junction or microscopic gap called a synapse
- The end of each axon are button like swelling called **terminal buttons** which contain small packets called **synaptic**

vessels where **neurotransmitters** are stored.

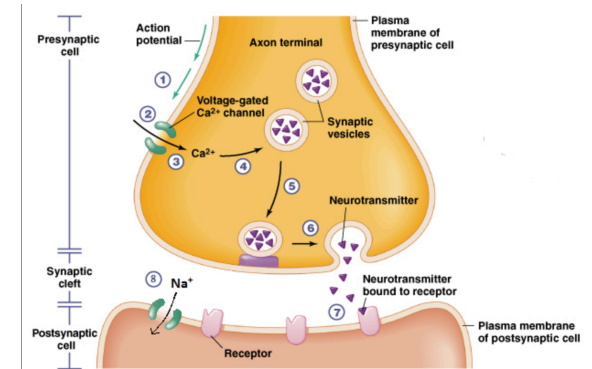


Figure 3. Synapses

NeuroTransmitter

- These are chemical substances released by one neuron through the synapse and affect or influence the receiving neuron.
- adequate number of neurotransmitters are released by the transmitting neuron, the receiving neuron will also fire.
- responsible for sensation, perception, cognitive and motor behavior. Excesses of or deficiencies of neurotransmitters have been linked to disease and abnormal behavior

TYPE	FUNCTION
Acetylcholine	- controls muscle contraction - It is found in the limbic system (forebrain) and is utilized by the motor neurons of the spinal cord to the muscles and glands of the body - Deficiency of ACH causes paralysis,

	forgetfulness and intellectual losses
Dopamine (<i>dopaminergic</i>)	involved in voluntary movements, learning, memory and emotional behavior
Serotonin	- Inhibitory transmitter which is involved in body temperature, sleep, moods, and pain sensation - Deficiencies cause anxiety, mood disorder, and insomnia - Foods rich in carbohydrates increase serotonin which leads to drowsiness in women and calmness in men
Endorphins	- Occur in the brain and bloodstream - Deficiency increases body pain - Excess (due to morphine, heroin, and other opiates) inhibit pain - Given post-operation to patients to reduce pain after anesthesia wears off
GABA (γ -aminobutyric acid)	- An inhibitory neurotransmitter that regulates the excitatory action of the neuron - A deficiency will result in anxiety Huntington's Disease: inherited disease is a

	result of loss of GABA-activated neurons resulting to inability to control the movement of the arms and legs with loss of cognitive capabilities Excess leads to excitation of neurons may result to epilepsy or an intense repetitive convulsive seizure
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Table 5. Five Types of Neurotransmitters

B. ENDOCRINE SYSTEM

Endocrine Glands

- These glands exert their control system by releasing chemicals called **hormones** into your blood.
- Hormones alter activity by affecting your nervous system like **neurotransmitters**

Catecholamines

- Composed of **Epinephrine (adrenaline)** and **Norepinephrine (noradrenalin)**
- Located in the **hindbrain**
- Involved in wakefulness, learning process, memory, eating, and regulation of moods
- Deficiency leads to depressive moods and excess will lead to arousal and wakefulness
- Their release is under the control of the **sympathetic nervous system**

C. THE BRAIN

What is the Brain?

- Weights about 3 lbs.
- Master organ of the body
- The seatbelt of self-awareness, the locus of intelligence, compassion, and creativity
- 2 PARTS OF THE BRAIN**
 - Lower Management:**
 - Hindbrain
 - Midbrain
 - Upper Management**
 - Forebrain

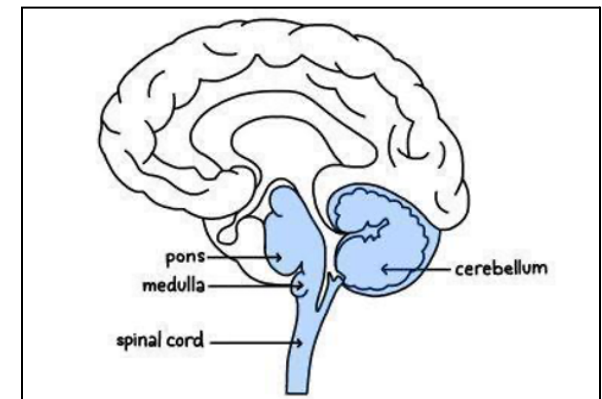


Figure 4. Parts of the Hindbrain and Midbrain

Lower Management

Hindbrain & Midbrain

- Spinal Cord**
 - Lowest level** of the central nervous system capable of making autonomic, reflexive "**management decisions**"
 - Management Decisions** processing inputs and generating outputs without direct brain involvement

- When you touch a stove, sensory-motor connections in your spinal cord withdraw your hand even before your higher levels of conscious awareness are notified.

- **Medulla**

- An enlargement of the brain stem involved in **regulating bodily functions** (e.g. breathing, heart rate, blood pressure) and certain reflexes.
- Also involved in **conveying messages to higher brain centers**.

- **Pons**

- A bulb-like structure at the top of the brain involved in **head movements** (e.g. chewing or swallowing) and **postural reflexes** (keeps your balance as you move or stand).
- Also involved in **conveying messages to higher brain centers**.

- **Cerebellum**

- Distinct ball-like structure behind the brainstem and below the cerebrum.
- It is involved in **coordination and balance**.
- *Thus, If there's a problem in your gait, then most probably there's a defect in your cerebellum.*

- **Midbrain**

- A small structure above the pons; that plays a role in **arousal, vision, and hearing**.
- Damage to this part leads to movement difficulties associated with **Parkinson's disease**

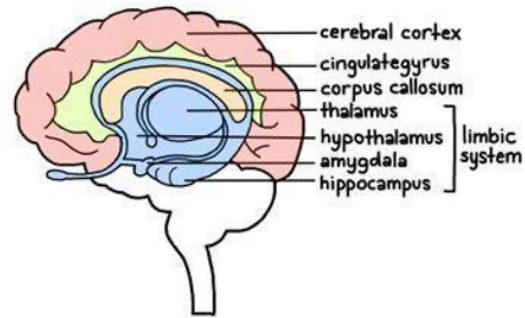


Figure 5. Parts of the Forebrain

Upper Management

Forebrain

- **Cortex**

- The thin outer layer of the brain that contains millions of nerve cells (neurons) that **influence most of what you think, feel, or do**.

- **Cerebrum**

- The cortical “peel” that covers the biggest part of the brain where most of the **important mental function** takes place (e.g. writing song lyrics, performing scientific experiments).

- **Thalamus**

- A structure deep within the brain that **conveys sensory information to the cerebrum** and other parts of the brain.

- **Limbic System**

- Composed of the
 1. **Hypothalamus**
 2. **Hippocampus**
 3. **Amygdala:** important in motivation and emotions

- The latter two—**hippocampus** and **amygdala**—are also important in memory.

III. HEALTH BEHAVIOR

Health Behavior

- Action taken by a person to maintain, attain, or regain good health and to prevent illness

A. RISK AND HEALTH

Risk

- **Oleckno (2002):** Probability of a specific event occurring in a given timeframe
- **Modifiable Risk**
 - Aspects of health in which an **individual has control of**
 - e.g. diet, smoking, sedentary lifestyle)
- **Non-modifiable Risk**
 - Aspects of health an **individual has no control of**
 - e.g. gender, genetic makeup, environmental exposures)

Risk Factors

- Exposure that is associated to disease

Risk Reduction

- **Proactive process**

- Behaviors that enable individuals to the actual actual/potential threats to their health

B. MAJOR FAMILY HEALTH RISKS

Biological and Age Related Risk

- Several illnesses are associated with either genetics or lifestyle patterns
- e.g. For cardiovascular diseases, it can often be traced through several generations of a family or can also be associated by the lifestyle of a family
 - Recent research findings support the positive effect of diet exercise and stress management in delaying cardiovascular diseases.
- e.g. For stress level theories, coping strategies can be identified to address the level of stress that the families would go through in sequence, if necessary, when families don't have an adequate variety of rules to follow to respond to an event, then the event will become stressful.

THE FAMILY STRESS THEORY OF LORENZ, WICKRAMA, AND CONGER (2004)	
Level 1	Change in the more specific patterns of behavior and transforming processes, such as change in who does which household chores
Level 2	Change in processes at higher level of abstraction, such as changes in what are considered

	as family chores
Level 3	Change in highly abstract processes, such as family value

Table 6. The Family Stress Theory of Lorenz, Wickrama, and Conger (2004)

Biological Health Risk Assessment

- One of the most effective techniques for assessing the patterns of health and illness in families is the genogram
- Genogram**
 - A drawing that shows the family unit of immediate interest and includes several generations using a series of circles, squares and connecting lines
 - Allows us to see hereditary patterns of behavior, medical and psychological patterns that run through families

TERM	DEFINITION
Genomics	Study of all the genes in the human genome and their interactions with other genes, the individual's environment, and the influence of cultural and psychosocial factors
	Social Risks: A family's health risk

Environmental Risks	increases if they are living in high crime areas, communities without adequate recreation or health resources, communities with noise pollution or chemical pollution Economic Risks: Determined by the relationship between the financial resources of a family and the demands on those resources
Behavioral (Lifestyle) Risk	Pattern of personal health habits and behavior defines individual and family lifestyle risk

Table 7. Biological Health Risk Assessment Definitions

C. BEHAVIORAL (LIFESTYLE) RISKS

Diet and Health

- Diet is one of the most modifiable risk factors**
- Obesity statistics in the Philippines: **7 out of 10 women; 1 out of 10 men** (due to periods, stress eating, etc.)
- Obesity affecting Filipino adults is known as the **android or apple-shaped type** where abdominal fat accumulation is measured using waist to hip ratio (WHR) recommended by WHO

- As research supports, the positive effect of diet, exercise and stress management delay the risk of developing cardiovascular disease
- An individual's body weight is determined by
 - Metabolism
 - Genetics
 - Behavior
 - Environment
 - Culture
 - Socioeconomic status
- **Metabolic Syndrome**
 - **Cluster of conditions** – increased BP, high blood sugar level, excess body fat
 - Typically have apple-shaped bodies (larger waists and carry a lot of weight around their abdomens)

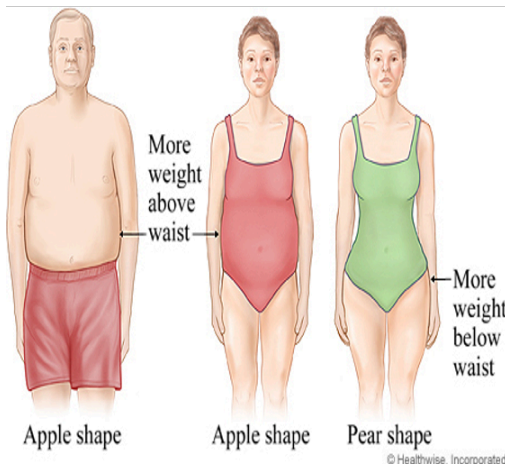


Figure 6. Image of Metabolic Syndrome
Physical Activity and Health

- Filipino adults are generally inactive (lazy)
 - 92.7% have low leisure related physical activity
 - 94.5% have low travel related activities (ex. Walking or cycling)

- 75.4% have low non-work related activities (ex. chores)
- 76.3% have low work-related activities
- The number of adults engaging in leisure activities with low or high intensity decreases with age
- Advantages:
 - Achieve weight management
 - Increased energy
 - Manage stress; Improve mood and self-esteem
 - Prevent development or worsening of a health condition
 - Better appearance
- Tool developed by **Clean Air Asia** to assess the suitability of walking in Asian cities.
- Parameters:
 - Availability of walking paths / crossing
 - Amenities (benches, pedestrians, trees, and street lights)
 - Disability infrastructure
 - Security from crime
 - Obstructions on walking pathways or pedestrians.
- “The first step is always the hardest.” Any form of exercise is better than none!

Sleep

- Insufficient sleep is associated with diabetes, heart diseases, obesity, and motor vehicle accidents.
- Sleep hygiene that will help achieve optimum sleep (National Sleep Foundation 2010)

- Avoid caffeine/nicotine and alcohol close to bedtime
- Finish exercising or vigorous activities at least 3 hours before sleep
- Retire and get up at the same time every day
- Avoid large meals before bed time

Postprandial Somnolence – normal state of drowsiness following a meal

- Establish a regular, relaxing bedtime routine (warm bath, etc.)
- Use the bed for sleep only

STAGE	AGE	SLEEP NEEDS
Newborns	1-2 months	10.5 to 18 hours
Infants	3-11 months	9 to 12 hours and 30 mins to 2 hours naps (1-4x a day)
Toddlers	1-3 years	12 to 14 hours
Preschoolers	3-5 years	11 to 13 hours
School-ages children	5-12 years	10 to 11 hours
Teens	11-17 years	8.5 to 9.25 hours
Adults and older adults	Above 17 years	7 to 9 hours

Table 8: Biological Health Risk Assessment Definitions

- The need of sleep is regulated by two processes:
 - **The number of hours we are awake**

- The extended hours we are awake should also be recovered by exceeding hours of sleep
- **Circadian biological clock in the brain**
 - How our brain responds to light
 - The circadian rhythm is why we are sleepiest between 2-4 am and between 1-3 pm

Psychoactive drugs

- Drugs that act on the CNS and alter its normal, everyday activity
- It causes changes in mood, awareness and behavior
- It disrupts the communication between neurons, so abusing them can have short term and long term effects on the brain

Categories:

- **Depressants**
 - Lowers the body's overall energy level, reduces sensitivity to outside stimulation and in high doses, induces sleep
 - Often decreases heart rate, respiration rate, muscular coordination and energy while dulling the senses
 - Examples: alcohol, barbiturates, benzodiazepines
- **Stimulants**
 - Activates or excites the nervous system – causes the nerve fibers to

release noradrenaline and other stimulating neurotransmitters

- These drugs don't give a person more energy but only spend its energy sooner and in greater quantities than they normally would
- Effect: sleepiness, laziness, mental fatigue, possibly depression (due to prevention of dopamine uptake)
- Examples: nicotine, cocaine, caffeine, amphetamines)
- **Hallucinogens**
 - Also called psychedelics (mind vision)
 - Can produce hallucinations, cause intoxication and lead to altered perception and impaired judgement
 - Physical effects are more constant, and it consists of CNS stimulation
 - Chronic use of these drugs can lead to the psychological effects and impaired judgement
 - They will mimic the effects of the neurotransmitters (serotonin and epinephrine), which lessens the pain
 - They can also trick the brain into seeing and hearing things that aren't actually there
 - Examples: LSD, psilocybin mushrooms and morning glory seeds
 - LSD - Lysergic acid diethylamide which

increases the production of neurotransmitters Gamma-Aminobutyric Acid (GABA) which decreases the reaction in the brain

- **Opioids**
 - Known to be narcotics
 - Substances that act on opioid receptors to produce morphine like effects
 - Examples: Heroin and pain medications

C. TOBACCO & HEALTH RISKS

- Smoking is the foremost preventable cause of death in the US
- In **2009, Philippines** ranked **Top 14 of Smoking Countries in the world**
- **Lung cancer comes second in the top 10 causes of cancer-related deaths** in the Philippines
 - First in rank is breast cancer, third for liver cancer, fourth for cervical, and fifth for colon
- Prevalence rate of Filipinos aged 15 years and older is 28%
 - 48% are males and 9% are females
- 3 categories of the primary causes of cigarette smoking related mortality
 - Cancer (Primary Lung)
 - Cardiovascular Diseases (Heart diseases and stroke)
 - Respiratory Diseases (Pneumonia, Influenza, Emphysema, etc.)
- Mainstream smoke
 - Can be inhaled directly by the smoker

- Side stream Smoke or Second Hand Smoke
 - Enters the atmosphere from the lighted end of the cigarette and inhaled by others

E. ALCOHOL CONSUMPTION & HEALTH RISKS

- (ethyl alcohol or ethanol)
- DRINK**
 - the amount of any alcoholic beverage delivering **½ an ounce of pure ethanol (around 15mL)** which is equivalent to the following:
 - 4-5 ounces (120-150mL) of wine.**
 - 12 ounces of beer (360mL) of beer.**
 - 1.25 ounces (40ml) of distilled liquor** (80 proof od whiskey, scotch, vodka, rum)

“normally, the liver can process about **½ ounce ethanol/hour**”

- Moderation** as defined by health authorities
 - Males:** not > 2 drinks/day;
 - Females:** not > 1 drink/day
- Age limit of purchasing alcohol in the Philippines is **18**.
- Drug-related death: When **alcohol is combined with HEROIN**. (Both depress the part of the brain that controls the basic functions of life—**breathing, body temperature, reflexes, blood pressure, and other vital functions.**)
- Types of drinkers:
 - Heavy drinking** - consuming more than 2 drinks/day (men), 1 drink/day (female) or 8 drinks/week (female) and 15 or more drinks/week (male)

- Binge drinking** - five or more (males) and 4 or more (female) drinks on a **single occasion**.
- Excessive drinking** - takes the form of heavy drinking, binge drinking, or both.

Short-term risks (due to binge/excessive drinking)	Long-term risks
Risky sexual behavior	Neurologic (stroke, dementia)
Violence	Myocardial infarction (MI), GI disturbance
Motor vehicle accidents	Hypertension
Falls/firearms/drowning	CNS problems - depression, anxiety, sleep, disturbance, memory loss
Miscarriage/stillborn	Cancer

Table 9: Alcohol Consumption & Health

- Facial Characteristics
 - Low Nasal bridge
 - Short nose
 - Indistinct Philtrum (groove between nose and upper lip)
 - Small head circumference
 - Small eye opening
 - Small midface
 - Thin upper lip

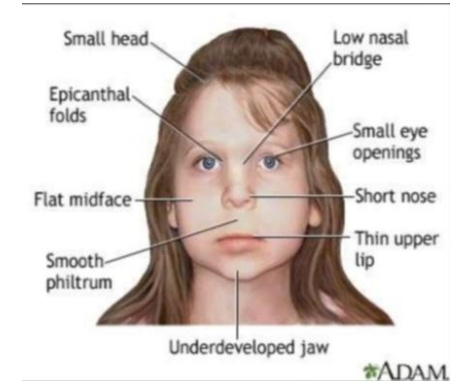


Figure 7. Image of Facial Characteristics

Blood Alcohol (%w/v)	Signs and symptoms
0.01 - 0.05	No obvious impairment, some changes may be observable.
0.03 - 0.12	Mild euphoria, Some impairment of motor skills
0.09 - 0.25	Loss of critical judgement, memory impairment, diminished reaction time
0.18 - 0.30	Mental confusion, dizziness, strongly impaired motor skills (walk unsteadily, slurred speech)
0.27 - 0.40	Unable to stand or walk, vomit, impaired consciousness
<u>0.35 - 0.50</u>	<u>Coma and possible death</u>

Table 10 Blood Alcohol Level

F. CAFFEINE INTAKE

- Most widely used psychoactive drugs in the world

- Found in coffee, tea, chocolates, soft drinks and various medications.
- Moderate doses (100 to 300 mg/day) increase mental alertness and probably have little negative effect on health
 - High doses can lead to insomnia, irritability, tremulousness, anxiety, cardiac dysrhythmia, GI disturbances and headaches.

A. HEALTH BELIEF MODEL

Health Belief Model

- Developed by a group of psychologists to explain why the public failed to participate in tuberculosis (Hochbaum, 1958)
- Through their work, this group found that information alone is rarely enough to motivate one to act – individuals must know what to do and how to do it before they can take action and the information must be related to one's needs
- It is based on the assumption that the major determinant of preventive health behavior is disease avoidance.
- **Limitation:** it assumes that only those clients who have distorted or negative perceptions of the disease or recommended health action will fail to act.

Concept	Definition
Perceived susceptibility	• One's belief regarding the chance of getting a given condition • Refers to a person's subjective perception of a risk of acquiring an illness or disease

	• There is a wide variation in a person's feelings of a person's personal vulnerability of a disease
Perceived severity	• One's belief regarding the seriousness of a given condition • One's belief regarding the seriousness of a given condition • Refers on a person's feelings on the seriousness of contracting an illness or disease • There is a wide variation in a person's feelings of severity and often a person considers the medical (death) or social (relationship with others) consequences
Perceived benefits	• One's belief in the ability of an advised action to reduce the health risk or seriousness of a given condition • Refers to a person's perception of the effectiveness of various actions available to reduce the threat of illness or disease • The course of action takes in preventing the illness or disease relies in the consideration and evaluation of both perceived susceptibility and benefit such that the

	person would accept the recommended action if it was perceived beneficial
Perceived barriers	• One's belief regarding the tangible and psychological costs of an advised action • Refers to a person's feelings on the obstacles to performing the recommended health action • There is a wide variation in a person's feelings of barriers or impediments which can lead to a cost or benefit analysis • The person weighs the effectiveness of an action against the perceptions that it may be expensive, dangerous or side effects that he/she may feel.
Cues to action	• Strategies or conditions in one's environment that activate readiness to take action • A stimulus is needed to trigger the decision making process to accept the recommended health action and these cues can be internal (wheezing pain) or external (asking advice from ppl who felt the same) or testimonies online in motivating yourself

Self - efficacy	- One's confidence in one's ability to take an action to reduce health risks - Refers to a person's level of confidence in his/her ability to successfully perform a behavior - Was added in the mid 1980's wherein many behavioral theories as it directly relates to whether a person performs the desired behaviors - A person decides whether he or not he continues the treatment

Table 10: Key Concepts and Definitions of the Health Belief Model

Transtheoretical Model

- Combines several theories of intervention, thus the name transtheoretical.
- based on the assumption that behavior change takes place overtime, progressing through a system of changes – one may stop in one stage, progress to the next stage, or return to the previous stage.
- Reasons why people resist change:**
 - Be unpleasant /stressful (ex. Exercising or eating new food)
 - Require giving up pleasure
 - Jeopardize social relationship
 - Require change in self image
 - Be painful
 - Not seem important anymore

Video: [The Transtheoretical Model of Change](#)

6 stages:	Characteristics:	Strategies:
Precontemplation	Denial ignorance of the problem Person has no intention of taking action	Encourage someone to rethink behavior Self-analysis/ introspection
Contemplation	Conflicted emotions or mixed-feelings Weighing pros and cons to change	Confirm readiness to change, and encourage confidence
Preparation	Person slowly collects information Has plans of action Experiment with small changes	Goals and give motivating statements
Action	Person takes direct action to achieve goal	Reward yourself with success Seek social support
Maintenance	Person has changed mindset already hence, avoid temptation and should continue to achieve desired behavior	Develops coping strategies Reward system

	For long-term already	
Relapse	Feeling disappointment or failure "What ifs"	Identify triggers Set barriers Remind yourself of your goals, COMMIT TO CHANGE

Table 10: Change Model or Transtheoretical model

The Theory of Planned Behavior & Theory of Reasoned Action

- Two Basic Assumptions that Underlie the TRA**
 - Behavior is under volitional control
 - People are rational being
 - From the perspective of TRA, we behave in a certain way because we choose to do so and we use a rational decision-making process in selecting and planning our actions

Video: [Theory of Planned Behaviour](#)

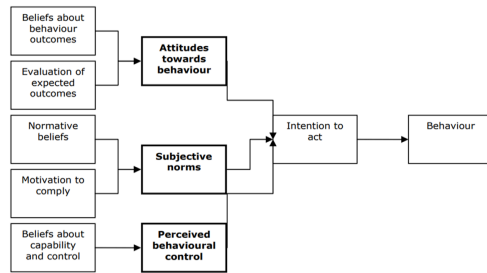


Figure 8: The Theory of Planned Behavior & Theory of Reasoned Action

Table 2. Theory of Reasoned Action/Planned Behavior	
Constructs	Description
Behavioral Intention	Perceived likelihood of performing the behavior
Attitudes	The product of the behavioral belief multiplied by the evaluation of it
Behavioral Belief	Evaluation of the likelihood that performance of the behavior is associated with certain outcomes
Evaluation of B.B.	How good or how bad those outcomes would be
Subjective Norm	The product of the normative belief multiplied by the motivation to comply
Normative Belief	Perception of how much each personal contact approves or disapproves of the behavior
Motivation to Comply	Motivation to do what each personal contact person wants
Perceived Behavioral Control	The product of the control belief multiplied by the perceived power
Control Belief	Perceived likelihood of each facilitating or constraining condition occurring
Perceived Power	Perceived effect of each condition in making the performance of the behavior easier or more difficult

Figure 9: Constructs of the Theory of Reasoned Action/Planned Behavior

Social Cognitive Theory

- The environment affects learning but learning outcomes depend on the learner's individual characteristics (Bandura, 1977)
- This theory goes well beyond individual factors in health behavior change to include environmental and social factors.
- In fact, this theory may be the most comprehensive model of human behavior yet proposed

- Based on the belief that learning takes place in a social context.
- Learning is promoted by modeling or observing other people.
- This theory means you always have someone to look up to
- It may be by giving detailed verbal instructions.
- Using print or multimedia strategies for health education
- What comes to our mind when we say "social" is the people surrounding you

Video: [Social Cognitive Theory](#)

V. COMMUNITY HEALTH EDUCATION

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- The process of changing people's knowledge, skills, and attitudes (KSA) for health promotion and risk reduction/disease prevention
- Empowering people to achieve optimum health and prevent disease by bringing about lifestyle changes and reducing exposure to health risks
- It is more expensive to get sick than to stay healthy
- e.g. teaching a pregnant woman on the need for regular prenatal consultations and instructing a family on methods of water purification that can be done at home.
- It is better to be safe than sorry.

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Community Health Education

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- Empowering people to achieve optimum health and prevent disease by **bringing about lifestyle changes** and reducing exposure to health risks
- A key to poverty alleviation. "It is more expensive to get sick than to stay healthy"
 - e.g. teaching a pregnant woman on the need for regular prenatal consultations and instructing a family on methods of water purification that can be done at home.
 - It is better to be safe than sorry.

A. SKILLS OF THE EFFECTIVE EDUCATOR

- The educator needs to understand the basic sequence of instruction

Steps that are Useful in Planning an Educational Program

1. Gaining the attention of the learners
2. Tell the learners the objectives of the instruction
3. Ask learners to recall previous knowledge related to the topic of interest so they will be able to link new knowledge with previous knowledge
4. Present the material in a clear, organized, and simple manner

5.
 - So that listeners can easily understand
6. Help learners apply the information to their lives and situation
 - You have to think of ways that listeners can relate
7. Encourage learners to demonstrate what they have learned
 - That is why there is RetDem or Return Demonstration, to see if learners have understood
8. Provide feedback to help learners improve their knowledge and skills

B. USES OF TECHNOLOGY IN HEALTH EDUCATION

- Now that we are in the modern days, readers prefer to read from the internet.

Criteria for Assessing the Quality of Internet Health Information

VanBiervlet and Edwards Schafer (2004)

- **Authorship**
 - Are the credentials and affiliations of authors and contributors listed?
- **Caveats**
 - Does the site clarify whether its function is to provide information or to market products?
- **Content**
 - Is the information accurate and complete, and is an appropriate disclaimer provided?
- **Credibility**
 - Does the site include the source, currency, relevance and editorial review process of the information?
- **Currency**

- Are dates listed for when the content was posted and updated?

- **Design**

- Is the site accessible, capable of internal searches, easy to navigate, and logically organized?

- **Disclosure**

- Is the user informed about the purpose of the site?

- **Interactivity**

- Does the site include feedback mechanisms and opportunities for users to exchange information?

- **Links**

- Have the links been evaluated according to back-linkages, content, and selection?

C. COMMUNITY HEALTH WORKERS (BARANGAY HEALTH WORKERS)

- Should be members of the community where they work
- They are selected by the communities and made answerable to the communities for their activities
- Supported by the health systems but not necessarily a part of its organization, and have shorter training than professional workers
- Provide primary care services—vital signs, physical assessment, and simple differential diagnostic techniques
- Must have communication, interpersonal, teaching, organizational, and advocacy skills; possess knowledge of their community, health issues and available resources
- They must also be trained in providing health services, such as basic assessment

and simple remedies for common health concerns

D. TWO-WAY REFERRAL SYSTEM

- A two-way referral system needs to be established between each level of health facility
- e.g. Barangay health workers refer cases to the rural health team, who in turn refer more serious cases to either the district hospital, then to the provincial, regional or the whole health care system.

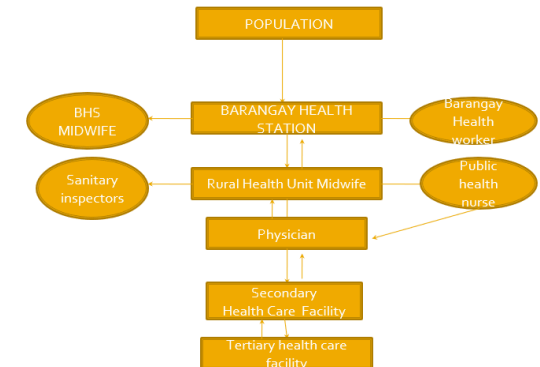


Figure 9: Diagram for the Two-Way Referral System

E. DIAGNOSTIC STAGES/PHASES OF EDUCATIONAL PLANNING FOR COMMUNITY HEALTH

Planning Phase

- **Organizing**
 - Activities
 - Appropriate methodologies, materials, resource person to conduct the topic, venue, send-out notice to the community members to inform them that you are

conducting a seminar in their community

- **Planning**
 - The content of health education and training activities to correspond to the knowledge, skills, and attitude (KSA) of the trainees of what they need to achieve
- **Identification of Tasks and Roles & Objectives of the Program**
 - Identify what “must be known,” “useful to know,” and “nice to know”

Implementation Phase

- Conduct of the **discussion/training, monitoring** (feedback/interactions), and **documentation** (highlights; reactions; issues/questions)
 - You have the program flow, where you start with the opening prayer, get to know the participants, ask them their expectations, give the overview of the seminar and also give the mechanics and house rules
- Opening prayer get-to-know participants expectations overview mechanics & house rules.
- Relate topic to be discussed to the role of the community health workers and to them, as community members
 - You have to ask and draw out the participants’ understanding of the topic, so you have to make sure they understood the topic.. Then, you have to summarize, reinforce and start the action planning.

Evaluation Phase

- Evaluation in the form of games, **return demonstrations**, or exercises to **establish participants’ KSA**, simple tests with options for trainees to open notes to reinforce learning
- **May be open to individuals and group evaluation**
 - Individual evaluation may be more effective because you can see if they really understood
 - Effective in identifying strengths & weaknesses of the training/discussion
- **CHW (Community Health Worker) performance evaluation** conducted by analyzing results of learning process and impact
 - **1st Phase:** CHW’s self-evaluation
 - **2nd Phase:** oral evaluation & skills assessment
 - **3rd Phase:** written examination

Post-Training Phase

- **Discuss results** of the evaluation—follow-up training activities in the future.
- Share the **identified strengths and weaknesses** in relation to activity
- Monitoring of trainees as they apply **their learning in actual life situations** by home or area-visits
- Encourage participants to **re-echo** what they learned **follow-up support** for trainees